Lecture 11: Node Voltage Analysis

EE201 – Circuit Theory I

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current voltage, node voltage e supernode

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalents
- Series, parallel
- Source transformation

Many circuits, not all

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- KCL: $A\mathbf{i} = \mathbf{0}$
- KVL: $\mathbf{v} = A^T\mathbf{e}$
- Element laws

Every circuit

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Only KCL is left

Motivation

Simplifying the circuit

- So far: simplify

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Node voltage analysis

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Motivation

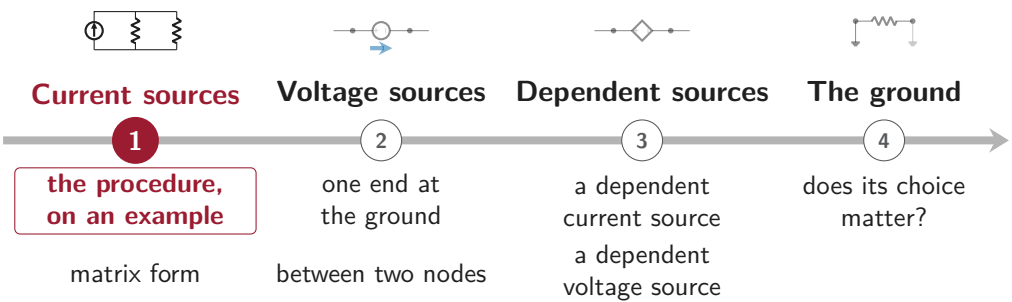
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Node voltage analysis

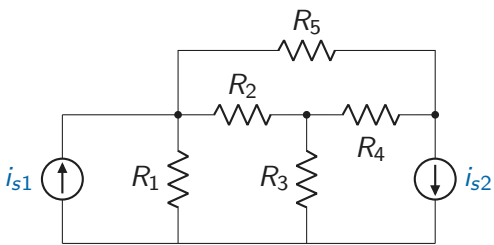
- Original circuit
- Unknowns: $n-1$ node voltages
- Then every v and i
- Every LTI resistive circuit

Roadmap


Current sources
Voltage sources
Dependent sources
The ground
1
2
3
4
the procedure, on an example
matrix form
one end at the ground
between two nodes
a dependent current source
a dependent voltage source
does its choice matter?

Example 11.1: node voltage analysis step by step

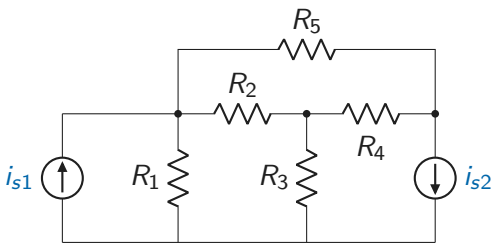
For now: only resistors and independent current sources



Question. How many unknowns?

Example 11.1: node voltage analysis step by step

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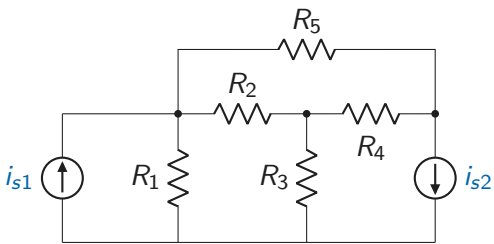


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$b = 7$: 14 unknowns

Example 11.1: node voltage analysis step by step

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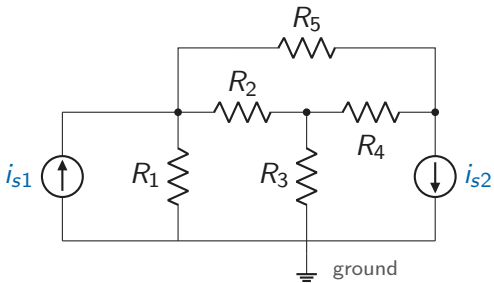


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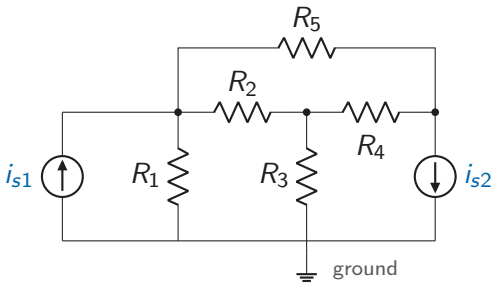
Node voltage analysis: 3

Step 1: choose a reference node



Reference node: the ground

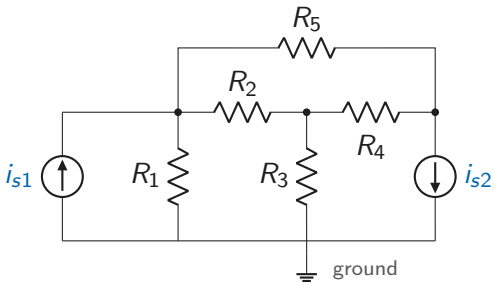
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Reference node: the ground

Its node voltage: 0

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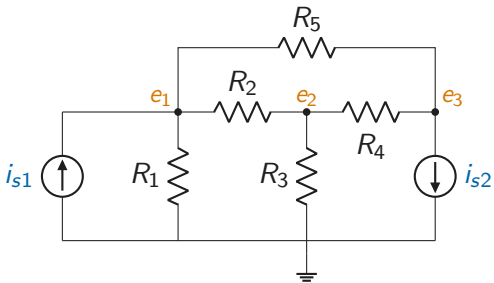


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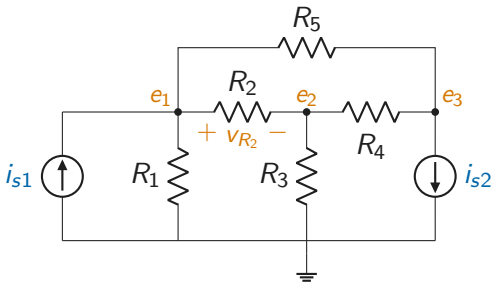
Any node can be the ground

Step 2: label the node voltages



Unknowns: e_1, e_2, e_3

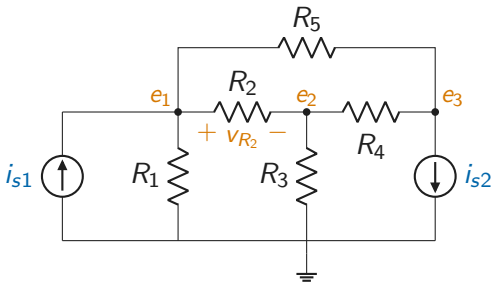
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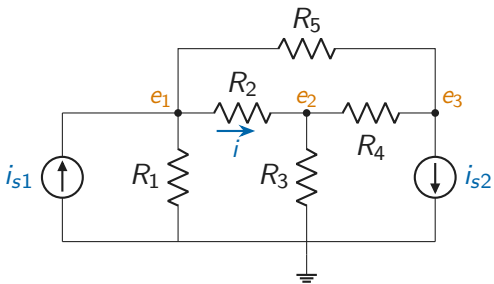
$v_{R_2} = e_1 - e_2$

KVL holds automatically:

$v_{R_1} = v_{R_2} + v_{R_3}$

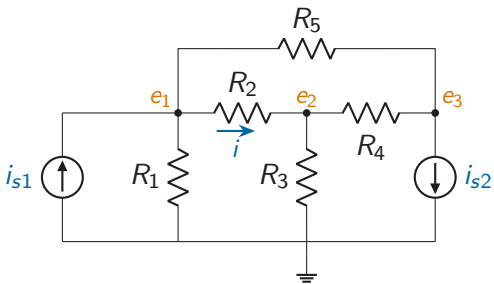
$e_1 = (e_1 - e_2) + e_2$

Step 3: write the currents with node voltages



Question. i leaving node 1 through R_2?

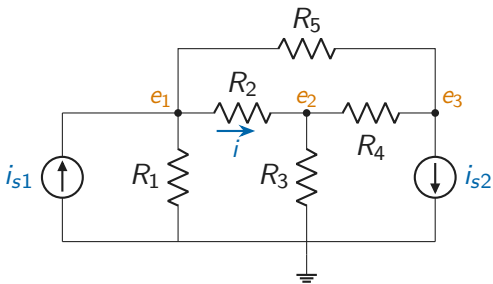
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$$i = \frac{e_1 - e_2}{R_2}$$

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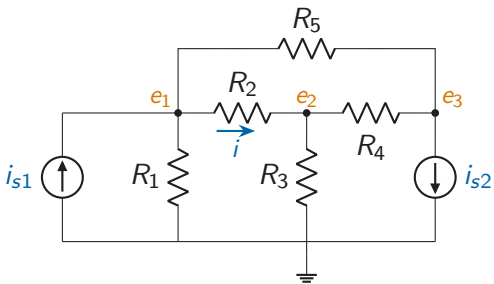


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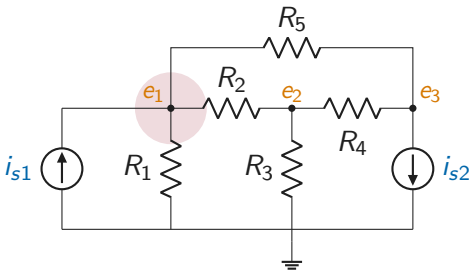
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$$\boxed{\frac{\text{own} - \text{other}}{R}}$$

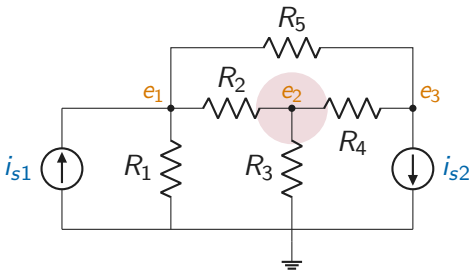
Step 4: KCL at every node except the ground



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

Step 4: KCL at every node except the ground



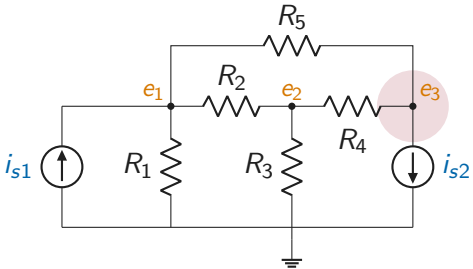
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Node 2:

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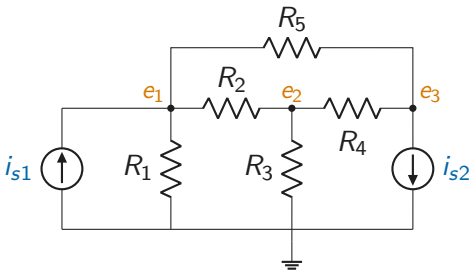
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Node 3:

$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_1}{R_5} + i_{s2} = 0$$

Step 4: KCL at every node except the ground



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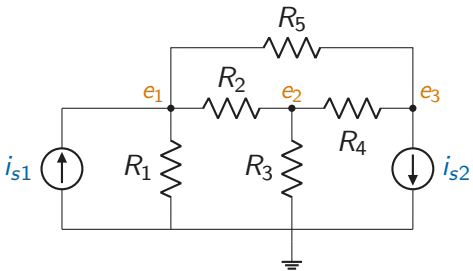
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Source into the node: minus

Step 4: KCL at every node except the ground



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Source into the node: minus

3 equations, 3 unknowns

Roadmap









Current sources | **Voltage sources** | **Dependent sources** | **The ground**

1 — 2 — 3 — 4

the procedure, on an example ✓

matrix form

one end at the ground

between two nodes

a dependent current source

a dependent voltage source

does its choice matter?

Example 11.1 in matrix form

$G_k = 1/R_k$:

$$
\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\
-G_5\, e_1 - G_4\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}
$$

Question. One matrix equation?

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Y_n: node admittance matrix

Example 11.1 in matrix form

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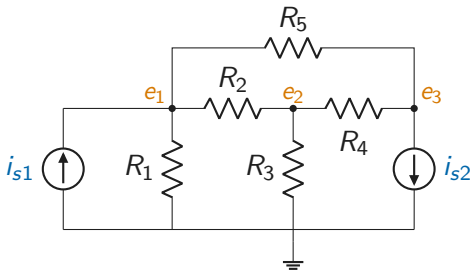
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$$

Y_n: node admittance matrix

$\mathbf{i}_n$: source currents entering each node

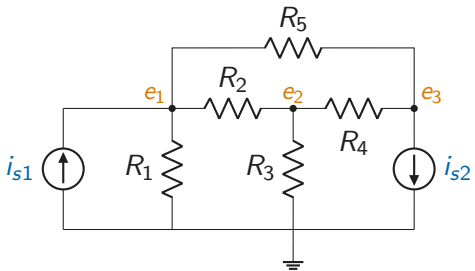
Writing Y_n directly from the circuit



$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

Question. Read Y_n from the circuit?

Writing Y_n directly from the circuit

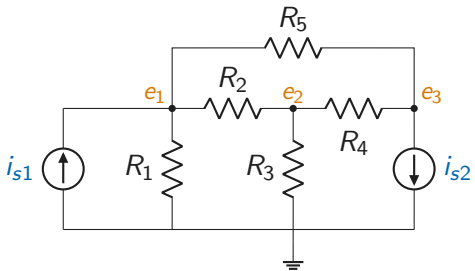


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Diagonal (k,k):
$+\sum G$ at node k

Writing Y_n directly from the circuit



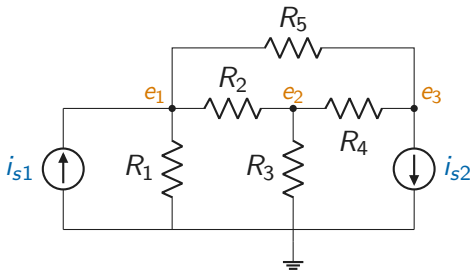
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Diagonal (k,k):
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Off-diagonal (k,j):
$-\sum G$ between k and j

Writing Y_n directly from the circuit



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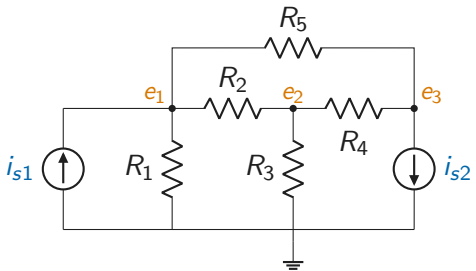
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Off-diagonal (k,j):
$-\sum G$ between k and j

Right side k:
$\sum i_s$ entering node k

Writing Y_n directly from the circuit



$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

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Y_n is symmetric

Roadmap









Current sources | **Voltage sources** | **Dependent sources** | **The ground**

✓ | 2 | 3 | 4

the procedure, on an example ✓

matrix form ✓

one end at the ground

between two nodes

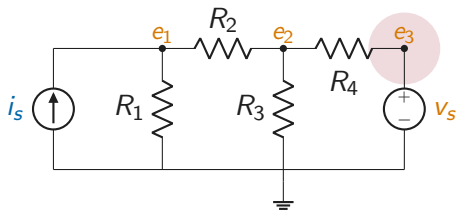
a dependent current source

a dependent voltage source

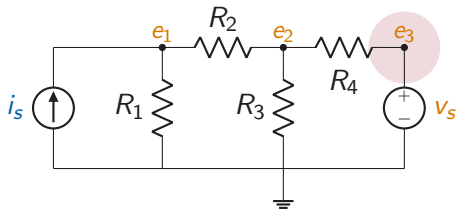
does its choice matter?

Example 11.2: a voltage source to the ground

$e_3 = v_s$: known



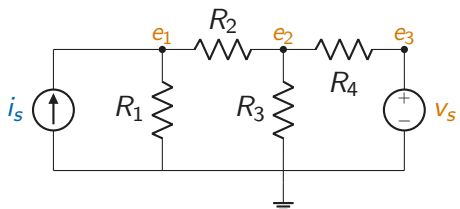
Example 11.2: a voltage source to the ground



$e_3 = v_s$: known

No KCL at node 3. Unknowns: e_1, e_2

Example 11.2: a voltage source to the ground

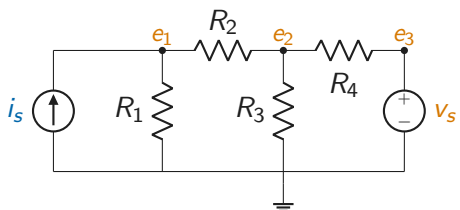


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No KCL at node 3. Unknowns: e_1, e_2

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Example 11.2: a voltage source to the ground



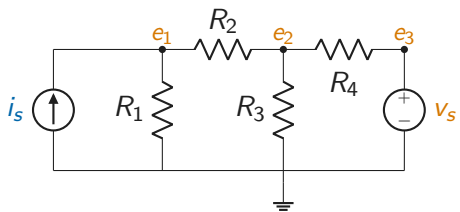
$e_3 = v_s$: known

No KCL at node 3. Unknowns: e_1, e_2

Node 1: $$-i_s + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} = 0$$

Node 2: $$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - v_s}{R_4} = 0$$

Example 11.2: a voltage source to the ground



$e_3 = v_s$: known

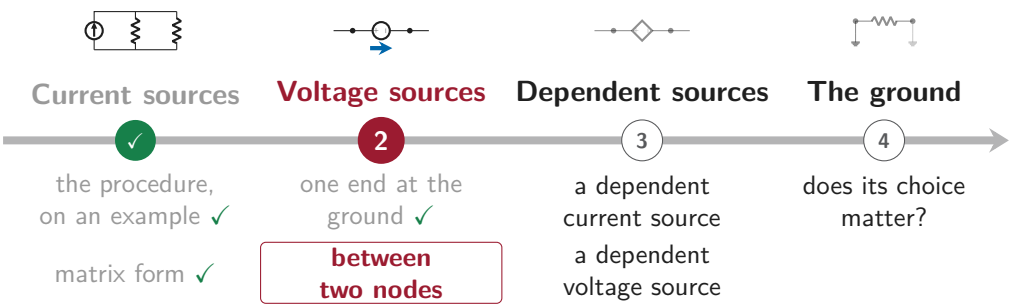
No KCL at node 3. Unknowns: e_1, e_2

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

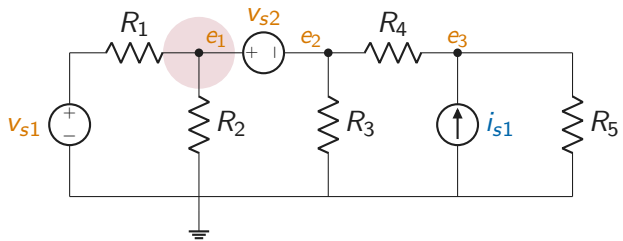
Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

Roadmap

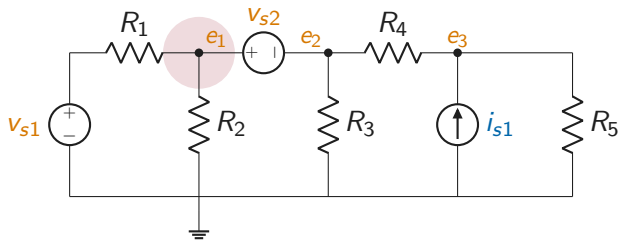

Current sources
Voltage sources
Dependent sources
The ground
✓
2
3
4
the procedure, on an example ✓
matrix form ✓
one end at the ground ✓
between two nodes
a dependent current source
a dependent voltage source
does its choice matter?

Example 11.3: a voltage source between two nodes



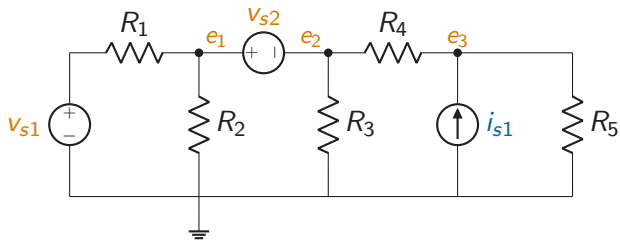
- **Question.** KCL at node 1: current through v_{s2}?

Example 11.3: a voltage source between two nodes



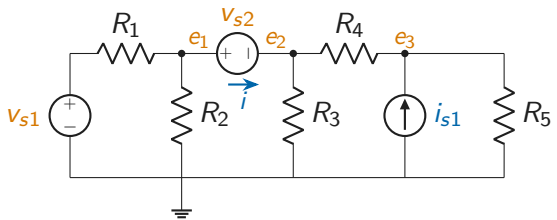
- **Question.** KCL at node 1: current through v_{s2}?
- Not given by e_1, e_2: a voltage source fixes v, not i

Example 11.3: a voltage source between two nodes



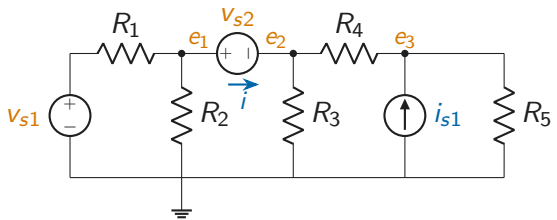
- **Question.** KCL at node 1: current through v_{s2}?
- Not given by e_1, e_2: a voltage source fixes v, not i
- Method I: its current as an unknown Method II: supernode

Method I: the current as an unknown



New unknown: i

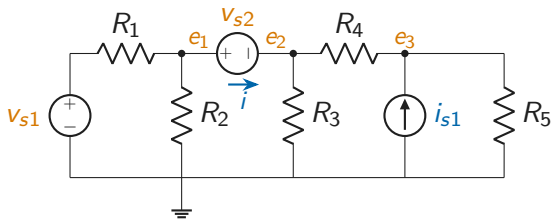
Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

New unknown: i

Method I: the current as an unknown

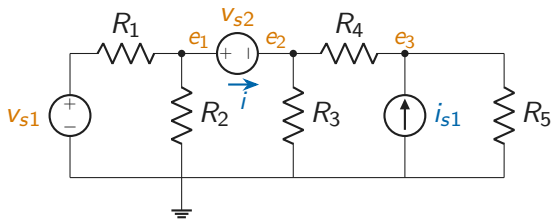


Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

New unknown: i

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Method I: the current as an unknown



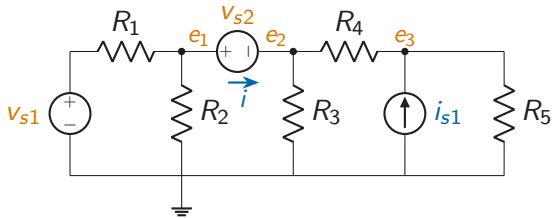
Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

New unknown: i

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

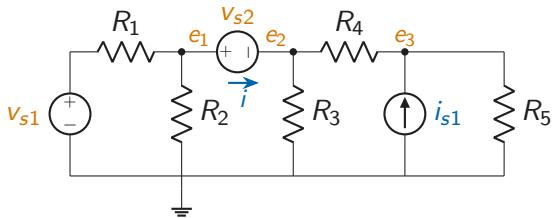
Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

New unknown: i

Constraint: $e_1 - e_2 = v_{s2}$

Method I: the current as an unknown



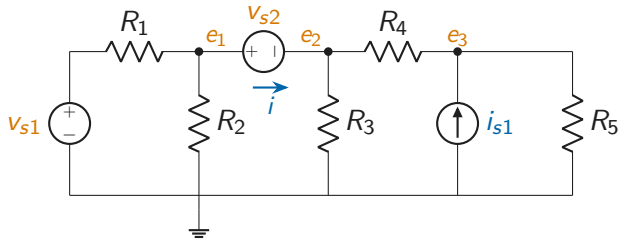
$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

New unknown: i

Constraint: $e_1 - e_2 = v_{s2}$

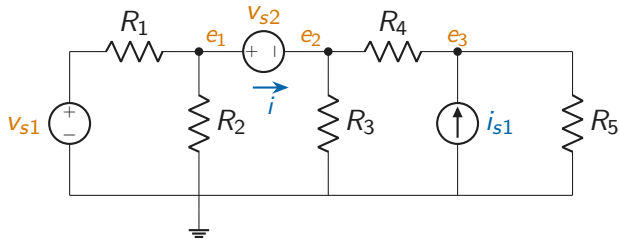
4 unknowns, 4 equations

Method II: the supernode



Question. Can we get rid of i?

Method II: the supernode

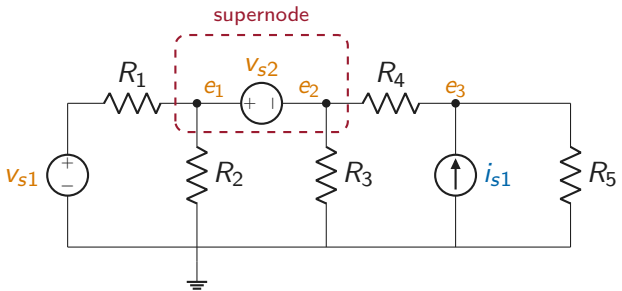


Question. Can we get rid of i?

Add KCL of nodes 1 and 2:

i cancels

Method II: the supernode

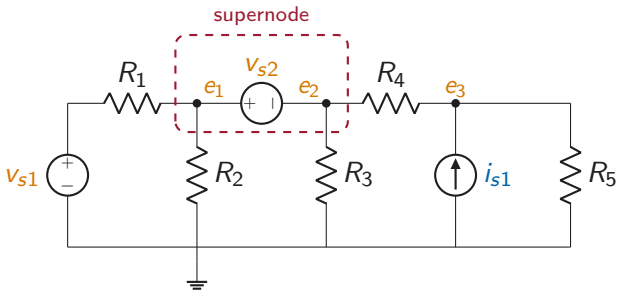


Question. Can we get rid of i? = KCL on a closed surface

Add KCL of nodes 1 and 2:

i cancels

Method II: the supernode



Question. Can we get rid of i?

Add KCL of nodes 1 and 2:

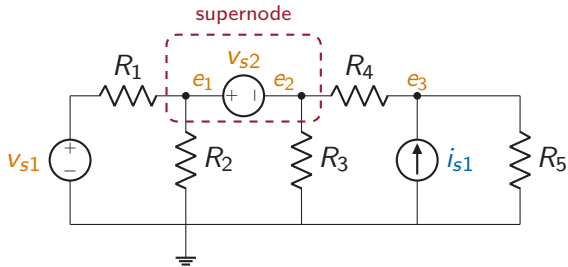
i cancels

= KCL on a closed surface

Supernode

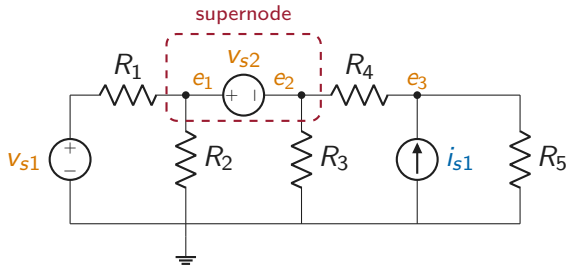
Two nodes of the source as one node

Method II: the equations



Question. KCL at the supernode?

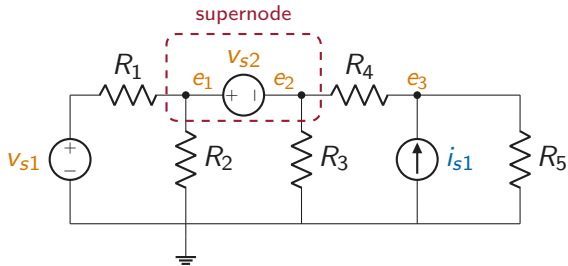
Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Method II: the equations

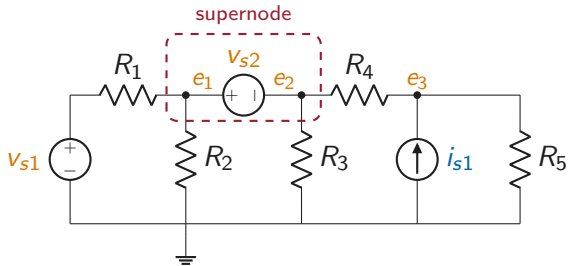


Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Method II: the equations



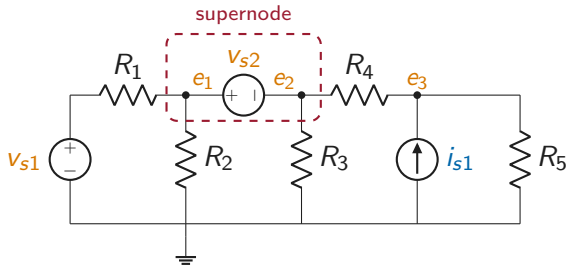
Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

Method II: the equations



Supernode:

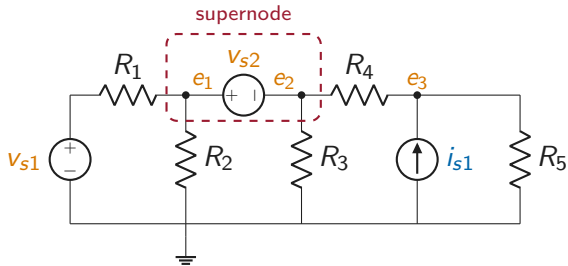
$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

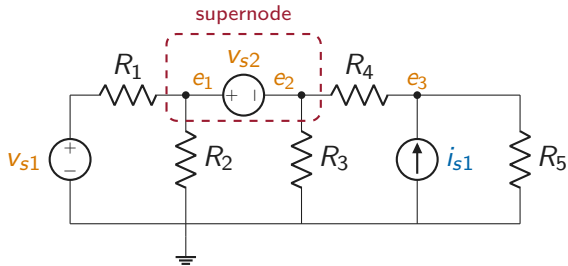
Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

3 unknowns (one fewer)

Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

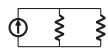
Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

3 unknowns (one fewer)

Not symmetric

Roadmap



Current sources

✓

the procedure, on an example ✓

matrix form ✓

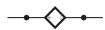


Voltage sources

✓

one end at the ground ✓

between two nodes ✓



Dependent sources

3

a dependent current source

a dependent voltage source



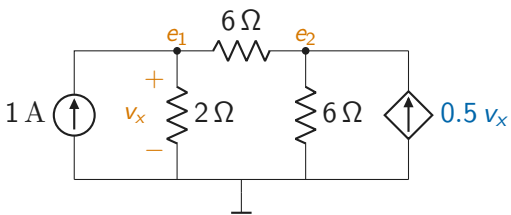
The ground

4

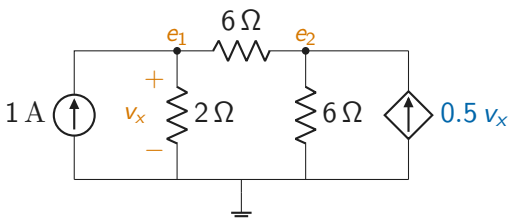
does its choice matter?

Example 11.4: a dependent current source

Question. Find e_1 and e_2.



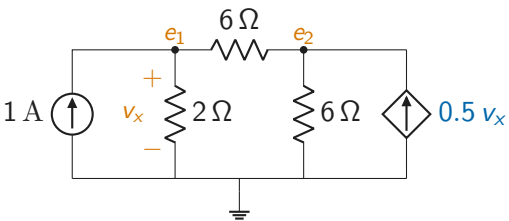
Example 11.4: a dependent current source



Question. Find e_1 and e_2.

Treat $0.5\,v_x$ as a current source

Example 11.4: a dependent current source

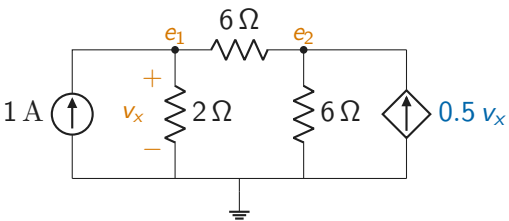


Question. Find e_1 and e_2.

Treat $0.5\,v_x$ as a current source

Node 1: $-1 + \dfrac{e_1}{2} + \dfrac{e_1 - e_2}{6} = 0$

Example 11.4: a dependent current source



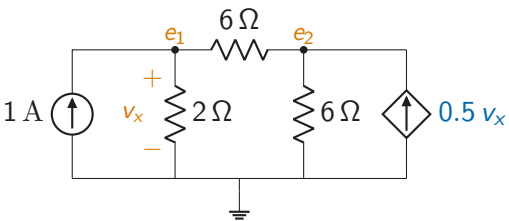
Question. Find e_1 and e_2.

Treat $0.5\,v_x$ as a current source

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Example 11.4: a dependent current source



Question. Find e_1 and e_2.

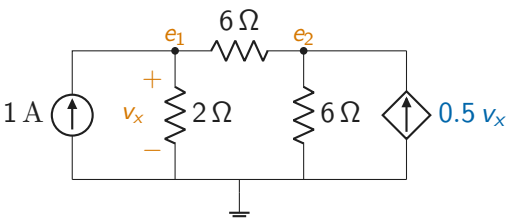
Treat $0.5\,v_x$ as a current source

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Question. v_x in node voltages?

Example 11.4: a dependent current source



Question. Find e_1 and e_2.

Treat $0.5\,v_x$ as a current source

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Question. v_x in node voltages?

$v_x = e_1$

Example 11.4: matrix form

First plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0$$

Example 11.4: matrix form

First plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

Example 11.4: matrix form

First plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Example 11.4: matrix form

First plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = 3\,\mathrm{V}, \quad e_2 = 6\,\mathrm{V}$$

Example 11.4: matrix form

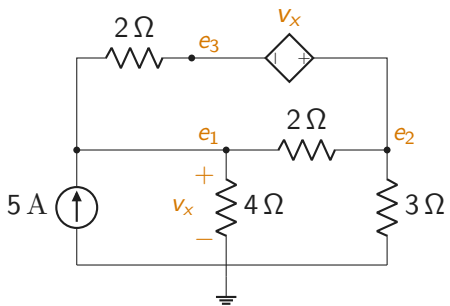
First plug $v_x = e_1$ into node 2:

$$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\, e_1 = 0 \qquad \Rightarrow \qquad -\frac{2}{3}\, e_1 + \frac{1}{3}\, e_2 = 0$$

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = 3\,\mathrm{V}, \quad e_2 = 6\,\mathrm{V}$$

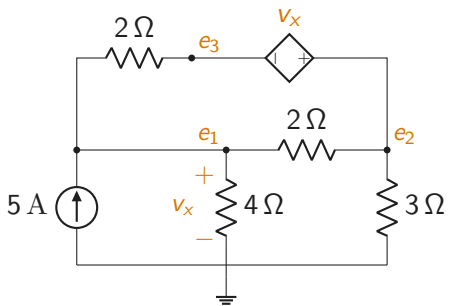
$-\frac{1}{6} \neq -\frac{2}{3}$: not symmetric

Example 11.5: a dependent voltage source



Question. Power absorbed by the 3 Ω?

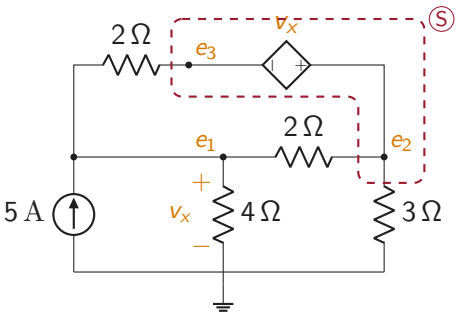
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v_x: between nodes 3 and 2, no ground

Example 11.5: a dependent voltage source

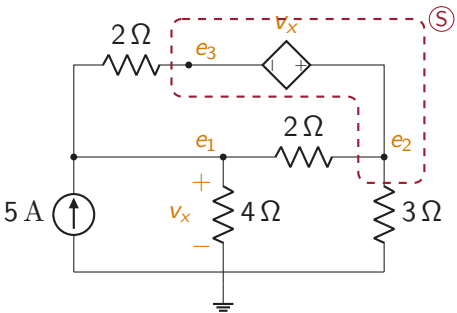


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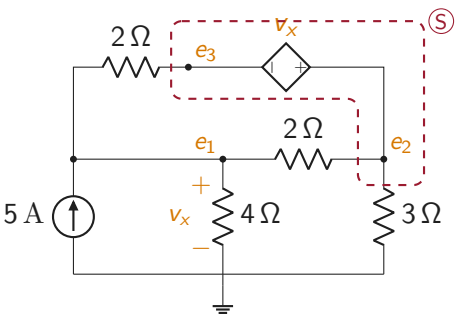
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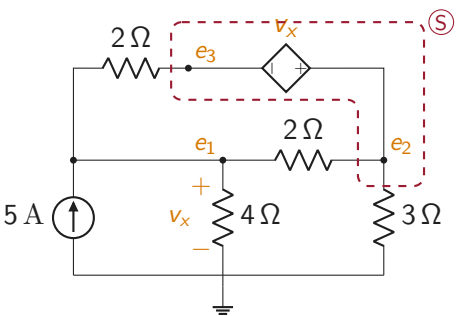
Unknowns: e_1, e_2, e_3

Example 11.5: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

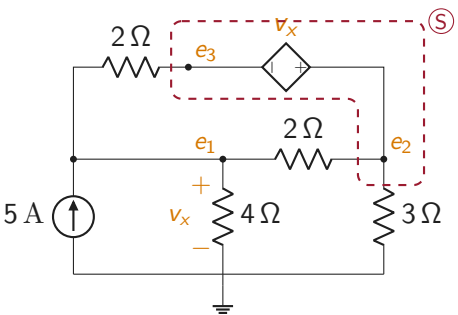
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Supernode Ⓢ: $$\frac{e_3 - e_1}{2} + \frac{e_2 - e_1}{2} + \frac{e_2}{3} = 0 \quad (2)$$

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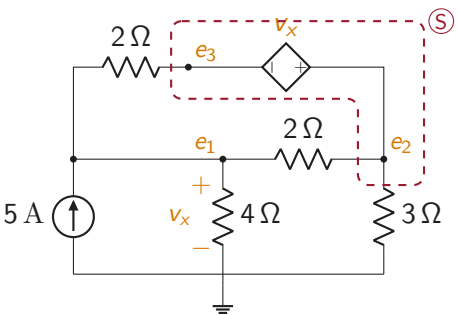


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Question. The constraint?

Example 11.5: the equations



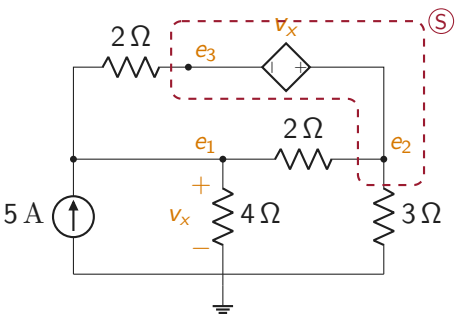
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Source: $e_2 - e_3 = v_x$

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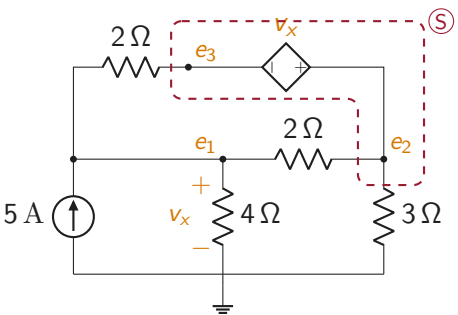
Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

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Source: $e_2 - e_3 = v_x$

Control: $v_x = e_1$

Example 11.5: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. The constraint?

Source: $e_2 - e_3 = v_x$

Control: $v_x = e_1$

Constraint: $e_2 - e_3 = e_1$ (3)

Example 11.5: matrix form

(1), (2), (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

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$$e_1 = 8\,\mathrm{V}, \quad e_2 = 9\,\mathrm{V}, \quad e_3 = 1\,\mathrm{V}$$

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$$e_1 = 8\,\mathrm{V}, \quad e_2 = 9\,\mathrm{V}, \quad e_3 = 1\,\mathrm{V}$$

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\mathrm{W}$$

Dependent sources

- Like an independent source of the same kind

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- Control variable in node voltages

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Remark

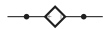
Dependent voltage source from node k to the ground:

no KCL at node k; write the constraint instead

Roadmap









Current sources ✓

the procedure, on an example ✓

matrix form ✓

Voltage sources ✓

one end at the ground ✓

between two nodes ✓

Dependent sources ✓

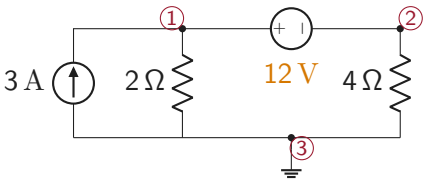
a dependent current source ✓

a dependent voltage source ✓

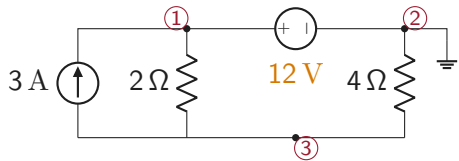
The ground 4

does its choice matter?

Example 11.6: where to put the ground?



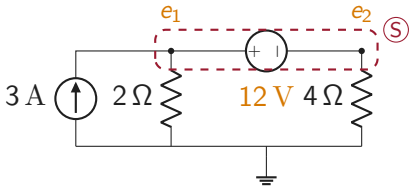
Ground at node 3

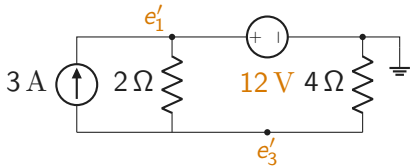


Ground at node 2: the − terminal of 12 V

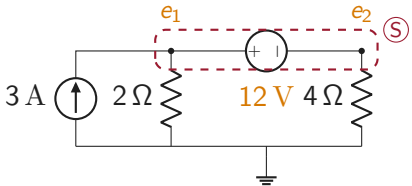
Question. Solve both. Which is easier?

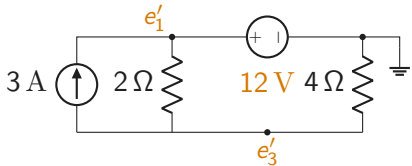
Example 11.6: side by side





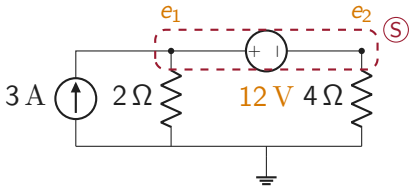
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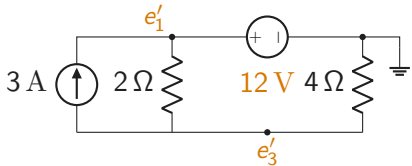




12 V between two nodes: supernode

Example 11.6: side by side

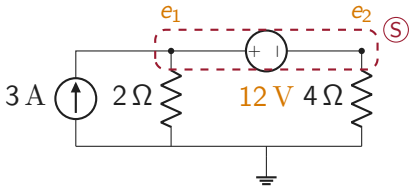


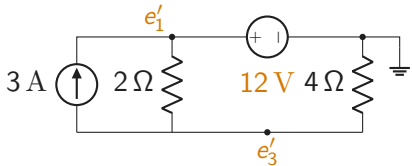


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S: $-3 + \frac{e_1}{2} + \frac{e_2}{4} = 0$

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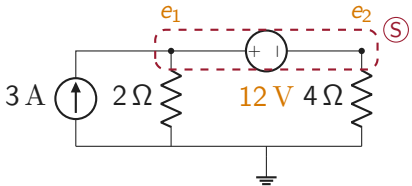


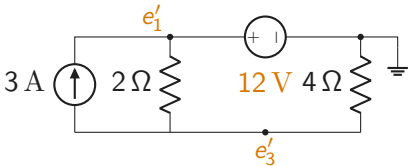
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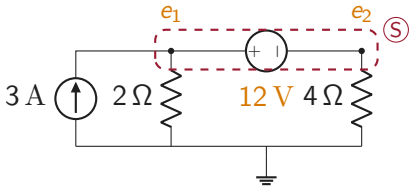
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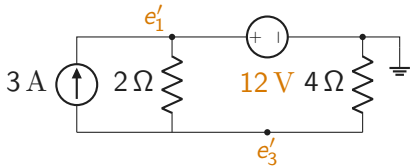


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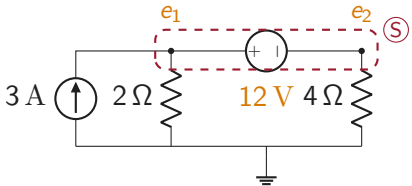
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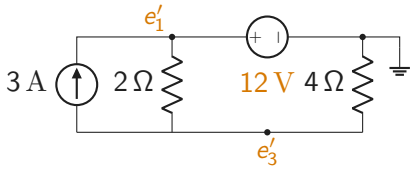


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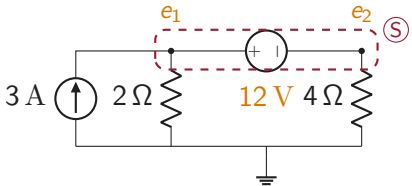
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Node 3: $3 + \frac{e_3' - 12}{2} + \frac{e_3'}{4} = 0$

Example 11.6: side by side

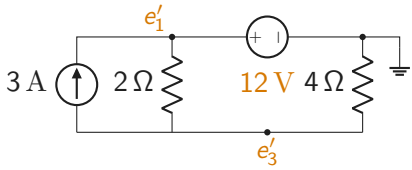


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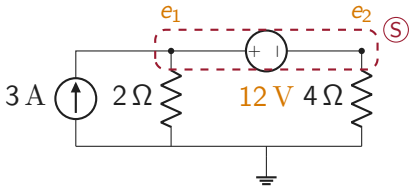


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$\frac{3}{4}\, e_3' = 3 \;\Rightarrow\; e_3' = 4\,\text{V}$

Example 11.6: side by side

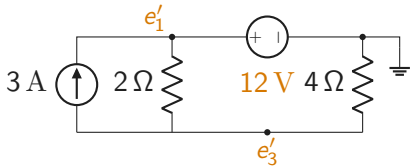


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2 unknowns | 1 unknown

Example 11.6: the same branch voltages

	ground at 3	ground at 2
$v_{2\Omega}$: node 1 to 3	$8 - 0 = 8$	$12 - 4 = 8$
$v_{4\Omega}$: node 2 to 3	$-4 - 0 = -4$	$0 - 4 = -4$

(V)

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Node voltages shift by $4\,\mathrm{V}$; branch voltages and currents do not change

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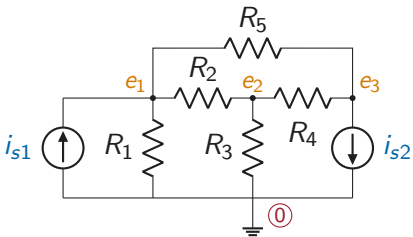
Tip. Put the ground at the $-$ terminal of a voltage source: one node voltage is known, and no supernode is needed

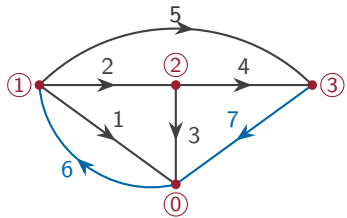
Summary

In the circuit	What to do
resistors, current sources	$Y_n\mathbf{e} = \mathbf{i}_n$, by the rules; symmetric
voltage source to the ground	e_k known; no KCL at node k
voltage source between nodes	Method I or supernode, plus constraint
dependent source	control in node voltages: constraint
choice of ground	any node; best at the $-$ terminal of a voltage source

Then $\mathbf{v} = A^T\mathbf{e}$ and the element laws

Y_n from the graph: the reduced incidence matrix

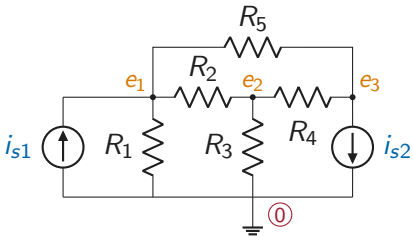


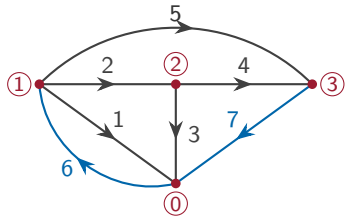


1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; ⓪: ground

Question. Reduced incidence matrix A?

Y_n from the graph: the reduced incidence matrix





1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; ⓪: ground

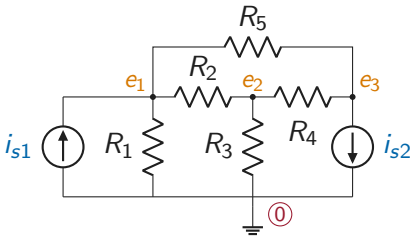
Question. Reduced incidence matrix A?

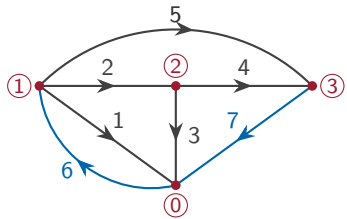
One row per non-ground node

$+1$ leaves, -1 enters

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccccc} & & & & & & \end{array}\right] \end{array}$$

Y_n from the graph: the reduced incidence matrix





1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; ⓪: ground

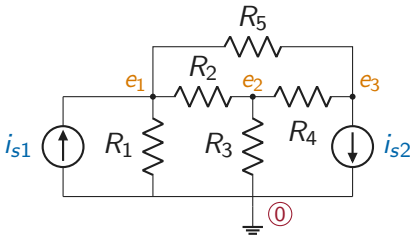
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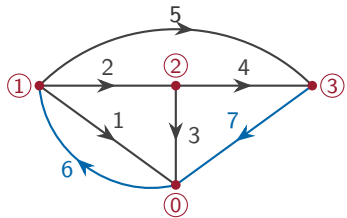
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$$A = \begin{matrix} \\ ① \\ ② \\ ③ \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ & & & & & & \\ & & & & & & \end{bmatrix} \end{matrix}$$

Y_n from the graph: the reduced incidence matrix





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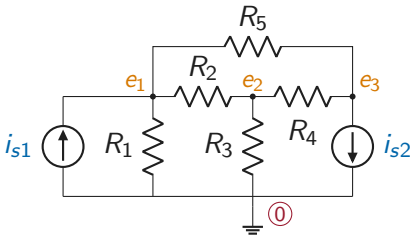
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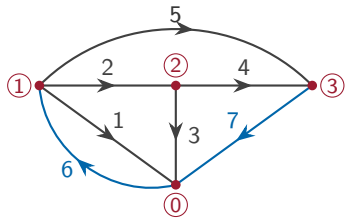
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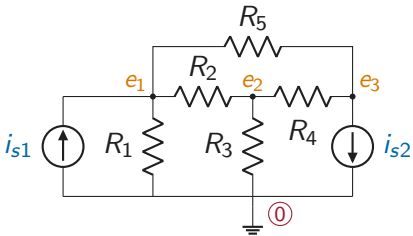
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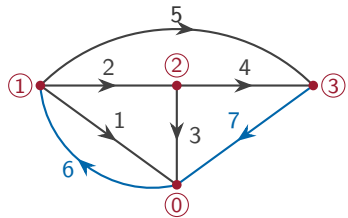
One row per non-ground node

$+1$ leaves, -1 enters

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

Y_n from the graph: the reduced incidence matrix





1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; ⓪: ground

Question. Reduced incidence matrix A?

One row per non-ground node

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A_R: resistors A_S: sources

Y_n from the graph: the general result

Resistors first, then current sources: $A = [\, A_R \;\; A_S \,]$

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$Y_n^T = Y_n$: always symmetric

Y_n from the graph: Example 11.1

From A: A_R = columns 1–5, A_S = columns 6, 7

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

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$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

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$$-A_S \mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix} = \mathbf{i}_n \quad \checkmark$$

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The rules = this product